

Percolation conditions in binary fractal fracture networks: Applications to rock fractures and active and seismogenic faults

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[1] The fracture network patterns in crystalline rock are treated using a binary fractal model described by three fractal geometric parameters: the fractal dimension D of the spatial distribution of fractures, the fractal dimension a of the fracture length distribution, and the maximum fracture length ($l_{\max}$) normalized by the domain length (L), $l_{\max}/L$. We present a percolation threshold relationship describing the boundary between percolating and nonpercolating conditions in random binary fractal fracture networks (RBFNs) and apply this to observations of natural fractures. Our results indicate that the three fractal parameters identified indeed control the connectivity of fracture networks, and hence the migration of fluids through rock fractures, and active and seismogenic faults. Moreover, our numerical approach may be applied in the future to investigating fluid migration associated with seismicity based on fractal parameters determined from earthquake catalog data. **INDEX TERMS:** 1848 Hydrology: Networks; 3250 Mathematical Geophysics: Fractals and multifractals; 5104 Physical Properties of Rocks: Fracture and flow; 8010 Structural Geology: Fractures and faults; 7230 Seismology: Seismicity and seismotectonics; **KEYWORDS:** fracture network pattern, fractal, percolation, seismicity, fluid migration

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1. Introduction

[2] One approach to understanding fluid flow and material transport through fractured crystalline rocks is to directly evaluate potential fluid migration pathways through the fracture network. Modeling the flow domain is extremely difficult, if not impossible, when the fracture network's effective permeability is more than 5 orders of magnitude higher than the intrinsic permeability of the rock matrix, since in these situations the groundwater flow paths are controlled solely by the fracture network's characteristics [e.g., Odling and Webman, 1991]. To realistically model such a network's hydraulic behavior, we must adequately represent the geometry of individual fractures, and then evaluate the connectivity of the network as a whole. The second step remains an unsolved problem in most realistic situations, despite being a vital objective in many civil engineering, environmental, and energy problems such as the geological disposal of high-level nuclear waste or the construction of artificial reservoirs by induced fracturing in hot dry rock power generation projects. An additional benefit of being able to accurately evaluate fracture network connectivity may be geophysical insight into the relationship between fluid migration and faulting activity at seismogenic depths [e.g., Tsuneishi and Nakamura, 1970; Byerlee, 1993; Tsunogai and Wakita, 1995; Miller et al., 1996; Yamashita, 1997].

[3] Many researchers have reported that fracture systems in crystalline rocks exhibit fractal spatial and fracture length distributions. For example, the spatial distribution of earthquake hypocenters has a fractal or multifractal geometry [e.g., Kagan and Knopoff, 1980; Kagan, 1981a, 1981b; Geilikman et al., 1990; Hirata and Imoto, 1991; Hirabayashi et al., 1992; De Rubeis et al., 1993; Godano and Caruso, 1995; Legrand et al., 1996; Godano et al., 1997; Nakaya and Hashimoto, 2002]. Moreover, seismogenic fault lengths are also found to follow a power law (fractal) distribution, as revealed by two empirical relationships: the Gutenberg-Richter frequency-magnitude relationship [e.g., Turcotte, 1997] and the relationship between magnitude M and faulting area S , $\log S = M - 4.07$ [Sato, 1979]. Several researchers have also concluded that the distributions of fracture aperture, throw, and spacing also have power law statistics [e.g., Marrett and Allmendinger, 1992; Gillespie et al., 1993; Bloomfield, 1996; Cladouhos and Marrett, 1996; Gillespie et al., 2001]. At smaller scales, fracture trace lengths in rocks also follow a power law distribution [e.g., Balberg et al., 1991; Gillespie et al., 1993; Bour and Davy, 1997; Van Dijk et al., 2000], as do the spatial distributions of fracture traces [e.g., Okubo and Aki, 1987; Hirata et al., 1987; Hirata, 1989b; Ohno and Kojima, 1992; Gillespie et al., 1993; Wilson, 2001].

[4] Overall, therefore, it has been observed that fracture network patterns in pervasively fractured crystalline rocks are typically characterized by two kinds of fractal, one concerning fracture lengths and the other the spatial distribution of fractures. A fracture network pattern described by such a binary fractal — that is, a fractal system consisting of

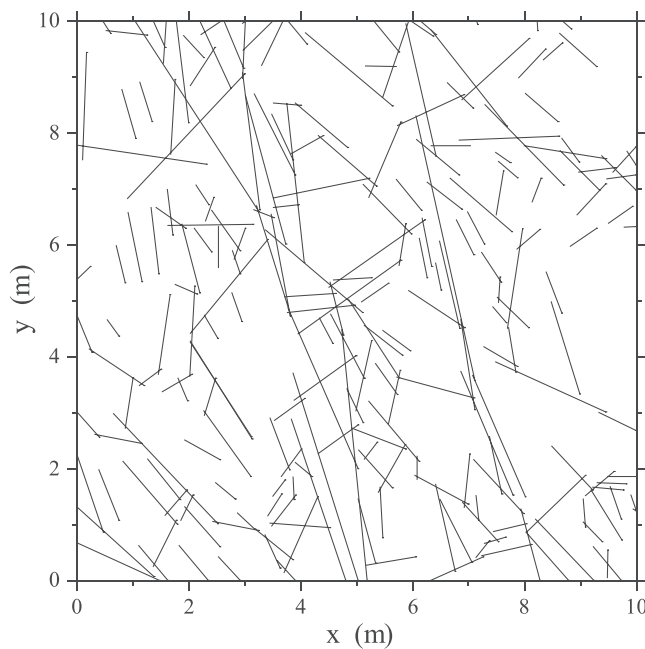


Figure 1. The sample fracture network pattern in straight lines (sample 4, traced from *Odling and Webman* [1991]).

two kinds of fractal geometries — can be described by several fractal parameters. Attempts to model fracture networks and the associated fluid flow in natural and synthetic systems have been made by *Martel et al.* [1991], *Balberg et al.* [1991], *Acuna and Yortsos* [1995], *Watanabe and Takahashi* [1995], *Bour and Davy* [1997], *de Dreuzy et al.* [2001a, 2001b], *Nakaya et al.* [2001], and *Bour et al.* [2002], among others. To date, however, little work has been performed on fluid flow in fracture networks described by binary fractals.

[5] In this study, the relationship between percolation probability and fractal parameters has been investigated

using a random two-dimensional network model, in order to understand fluid migration through a binary fractal fracture network. The geometries and fractal parameters of natural fracture networks in rocks of various scales were measured and used as inputs. The first goal of this study is to define the percolation condition equation (which describes the boundary between the percolated and nonpercolated domains in fracture networks) in terms of the binary fractal model. The second goal is to demonstrate the applicability of the binary fractal model at different scales. Finally, we have also evaluated the applicability of the binary fractal percolation equation to models of induced seismicity caused by fluid migration along faults.

2. Method

2.1. Fractal Modeling of Natural Fracture Network Patterns in Rocks

[6] The fractal model provides a useful model of the spatial distribution of fractures and the frequency distribution of fracture lengths. In order to model a naturally occurring fracture network in two dimensions, the fractal properties of the fractures comprising the network must be computed from measurements. The procedure used here is as follows: first, the coordinates of both ends of straight fracture trace segments are measured. Second, the fractal dimension D is computed using equation (1) for fractures within an $L \times L$ square domain using the box-counting approach [*Barton and Larsen*, 1985]. In equation (1), L is the domain length, $r = L/k$ ($k = 1, 2, 3, \dots$), and $N(r)$ is the number of boxes of size r that include the center points of fractures:

$$D = \lim_{r \rightarrow 0} \log N(r) / \log(1/r). \quad (1)$$

Third, the sampled fracture patterns are plotted as the logarithm of the cumulative frequency versus the logarithm

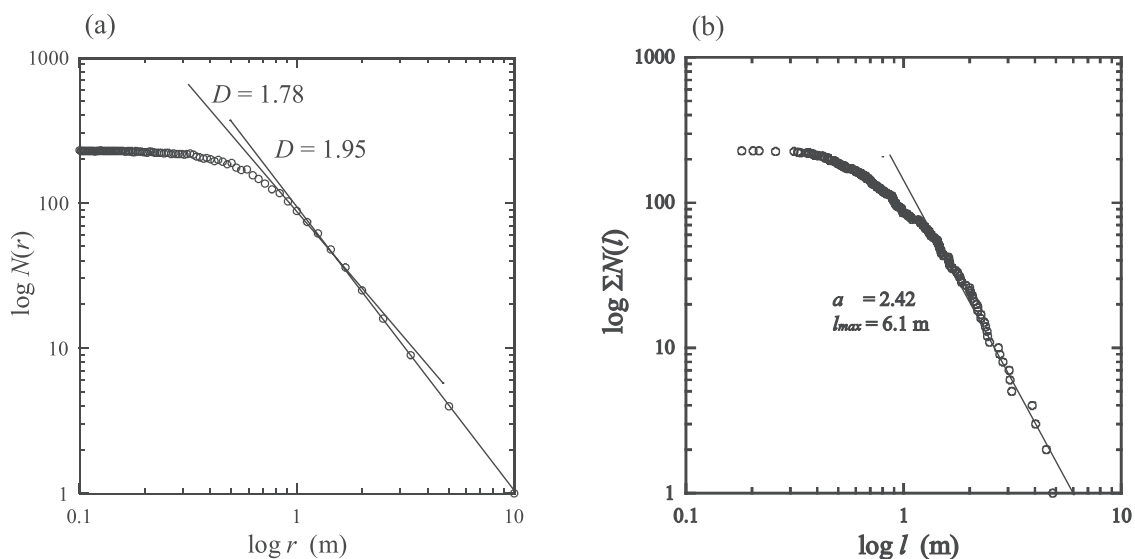


Figure 2. (a) $\log N(r) - \log r$ plot for spatial distribution of fracture centers and (b) $\log \sum N(l) - \log l$ plot for fracture length in fracture network pattern of sandstone (sample 4, analyzed from *Odling and Webman* [1991]; see Figure 1).

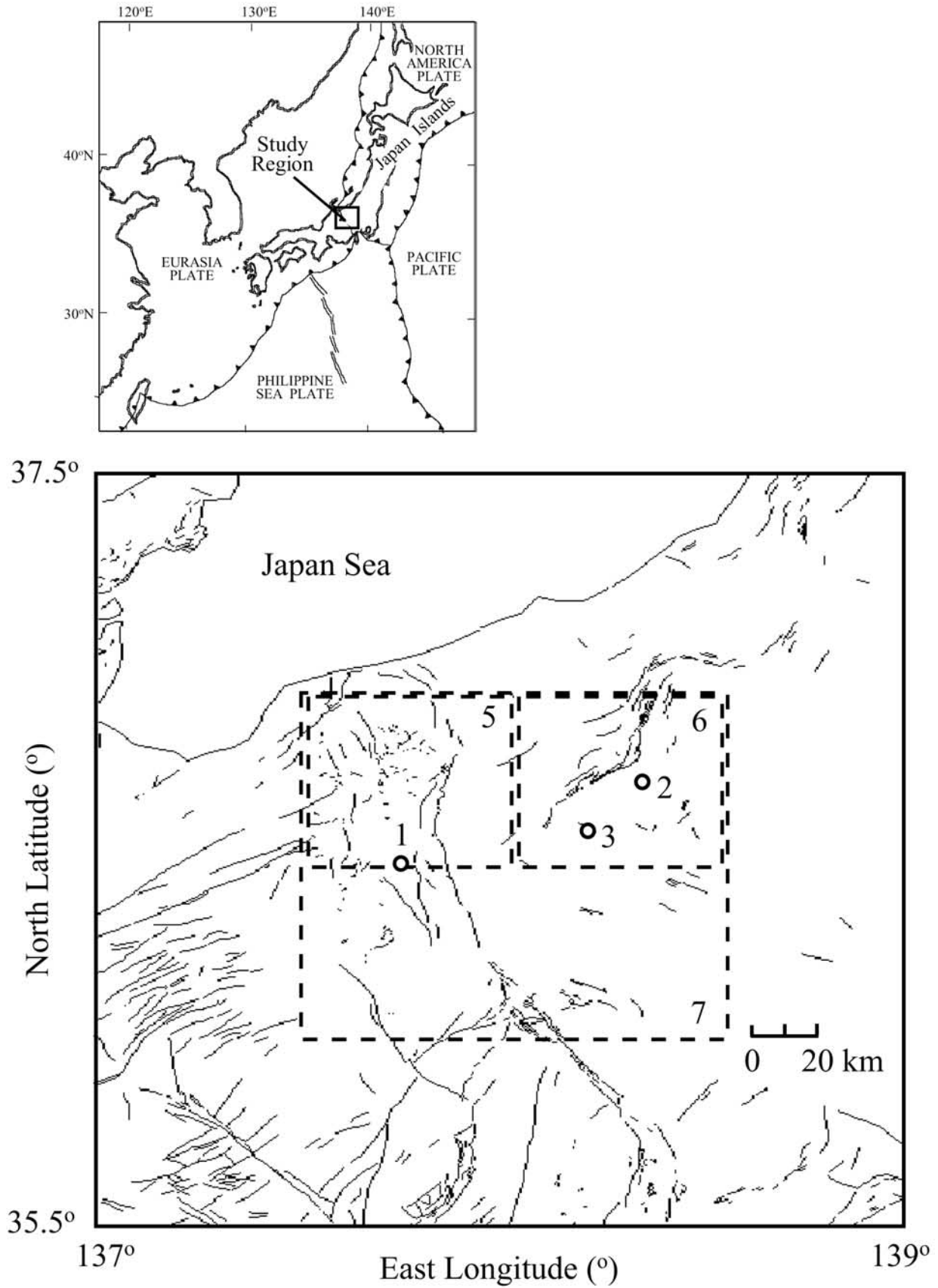


Figure 3. The locality of fracture sampling points. Faults are traced from *Research Group for Active Faults in Japan* [1991].

Table 1. Locality of the Sampling Points and the Domain Sizes of Fracture Samples

No.	Sample	Locality	Domain Size ($L \times L$)
1	granite	Kuzu/Nanakura	6.5 cm $\times$ 6.5 cm
2	andesite	Takayama Village	75 cm $\times$ 75 cm
3	green tuff	Matsushiro	6 m $\times$ 6 m
4 ^a	sandstone	Norway	10 m $\times$ 10 m
5 ^b	active fault 1	Nagano Prefecture	50 km $\times$ 50 km
6 ^b	active fault 2	Nagano Prefecture	50 km $\times$ 50 km
7 ^b	active fault 3	Nagano Prefecture	100 km $\times$ 100 km

^aFractures are traced from *Odling and Webman* [1991].^bFaults are traced from *Research Group for Active Faults in Japan* [1991].

of the trace length to determine the two fractal parameters of interest. Here a and $l_{\max}$ describe the power law

$$\Sigma N(l) = Bl^{-a}, \quad (2)$$

$$a = \lim_{l \rightarrow 0} \log \Sigma N(l) / \log(1/l),$$

$$B = (l_{\max})^a,$$

where $\Sigma N(l)$ is the number of fractures with lengths larger than l , and $l = l_{\max}$ when $\Sigma N(l) = 1$ within an $L \times L$ square

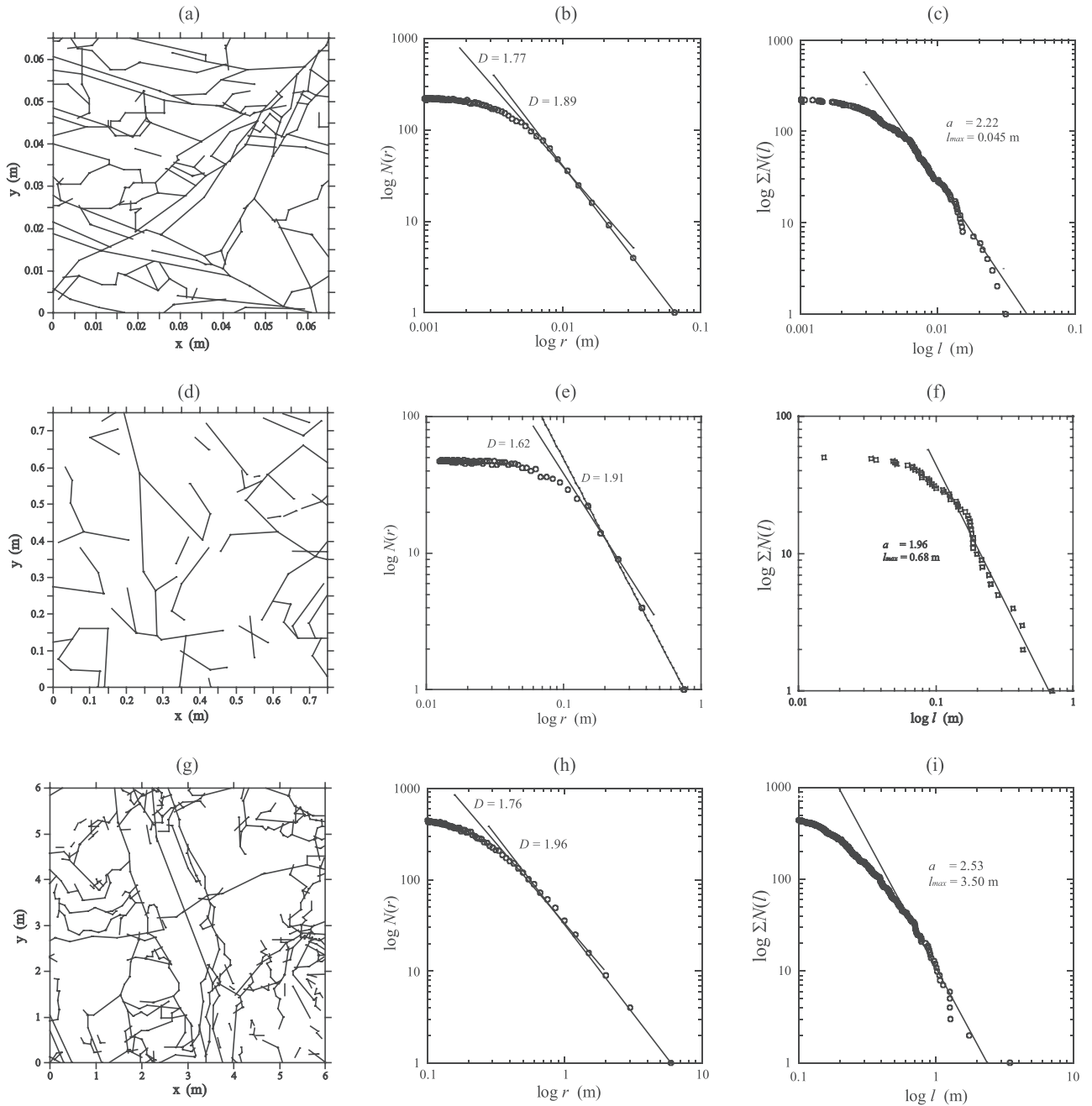


Figure 4. The linearized fracture network patterns, $\log N(r) - \log r$ plots and $\log \Sigma N(l) - \log l$ plots of samples. (a)–(c) Granite (1), (d)–(f) andesite (2), (g)–(i) green tuff (3), (j)–(l) active faults (5), (m)–(o) active faults (6), (p)–(r) active faults (7). Faults (Figures 4j, 4m, and 4p) are traced from *Research Group for Active Faults in Japan* [1991].

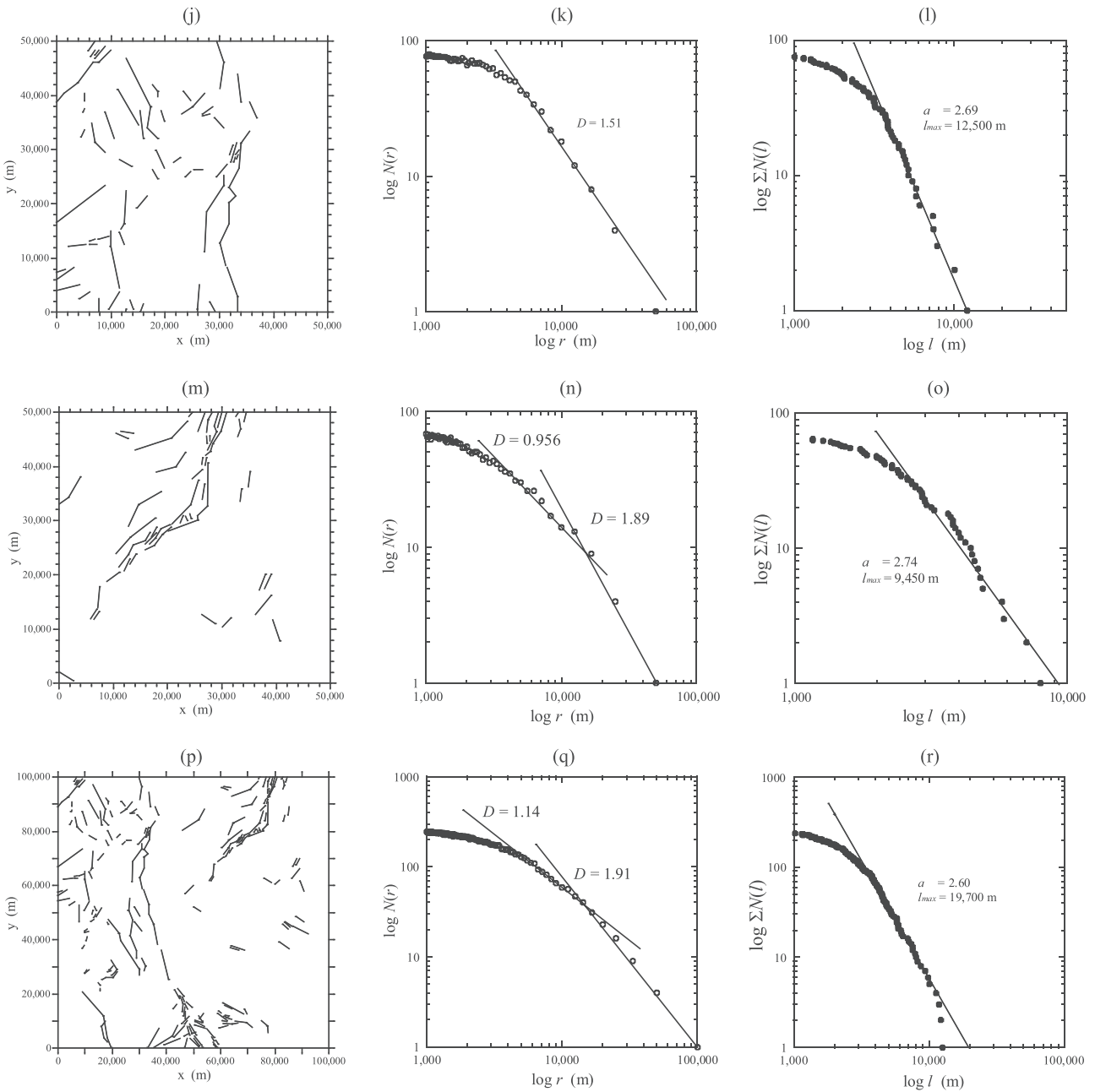


Figure 4. (continued)

domain. That is, a and $l_{\max}$ correspond to the negative slope of the regression line and the value of l at $\Sigma N(l) = 1$, respectively, in the $\log \Sigma N(l) - \log l$ plot, and are referred to as the fractal dimension of fracture length distribution and the maximum fracture length.

[7] The following four points should be borne in mind when determining a and D from $\log \Sigma N(l)$ versus $\log l$ plots and $\log N$ versus $\log r$ plots, respectively: (1) the error associated with representing fractures as straight lines on a two-dimensional plane, (2) the error resulting from long fractures extending beyond the sampling window, (3) the fractal limit of the fractures, and (4) errors associated with the box-counting method.

[8] The effect of point 1 is to shift the data points in the $\log \Sigma N(l) - \log l$ plot to the left as a single long, possibly

undulating fracture is divided into a number of smaller straight segments.

[9] As for point 2, any fractures that are too long to be completely contained within a scan window will be incorrectly represented as shorter fractures. Consequently, the data points in the $\log \Sigma N(l) - \log l$ plot fall at smaller values of l and $l_{\max}$ may be underestimated.

[10] As suggested by point 3, the fault network may not be truly fractal or otherwise possess only range-limited fractal properties [Walsh and Watterson, 1993; Wilson, 2001; Oncel et al., 2001]. Wilson [2001] reported that two linear regions separated by an abrupt transition are observed in the $\log N/\log r$ box curve in many cases, with the transition between the two regimes being related to the dominant spacing of individual elements or to the dominant

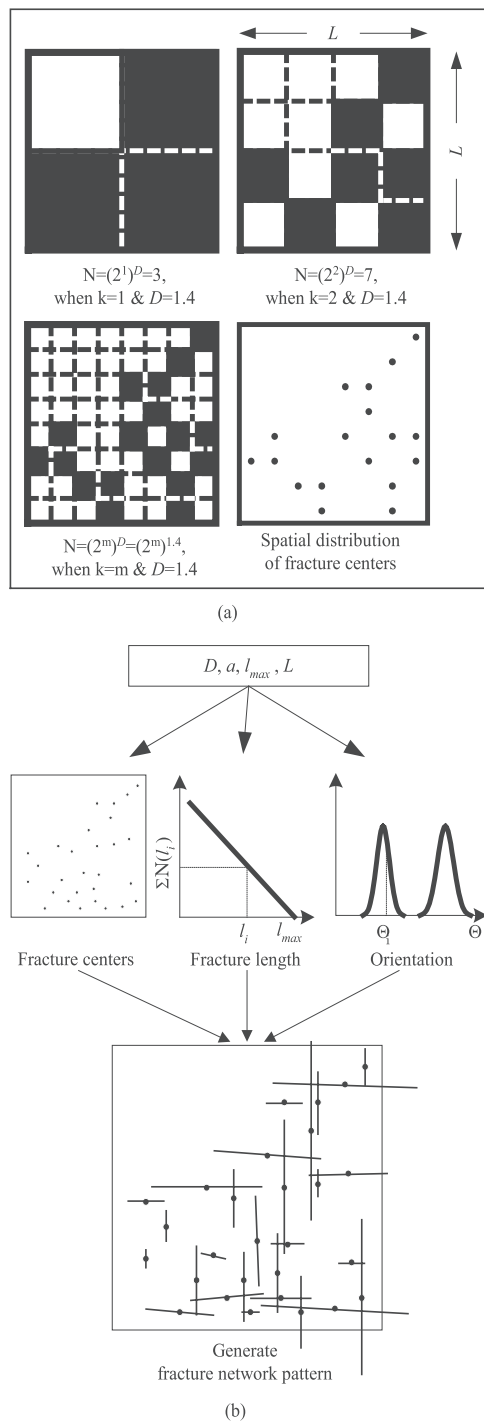


Figure 5. Generation procedure of a random binary fractal fracture network (RBFFN) model. (a) Fractal fragmentation with fractal dimension D for spatial distribution of fracture centers. (b) Generation of fracture network pattern with fractal parameters D , a , and l_{max}/L .

fragment size in the network. Wallace [1973] obtained a power law distribution with an exponent and an exponential cutoff to an upper fractal limit [Scholz, 1997, 1998] and suggested that the l value of this upper limit may bear a close relation to the thickness of the seismogenic zone.

Consequently, the data points in the $\log \Sigma N(l) - \log l$ plot may fall at smaller values of l .

[11] Finally, the effect of point 4 is related to the fact that large scanning windows are often occupied by only part of the fracture pattern [e.g., Walsh and Watterson, 1993; Wilson, 2001]. Consequently, estimates of slope from the high r end of the plot almost invariably yield anomalously high values and thus correspondingly high fractal dimensions; that is, the log-log response is typically curvilinear rather than strictly linear, with steepest slopes for high values of r .

[12] Taking into account these four potential limitations to the fractal parameter estimates, the following criteria are employed in this study: in cases in which two linear regions separated by an abrupt transition are observed in the $\log N/l$ vs $\log r$ box curves, two fractal dimensions D are determined. In estimating a , only the linear sections of the curve below some upper limit are used; l_{max} corresponds to the maximum length in the scan window in the case of point 1 and to l at the intersection point of the regression line and $\Sigma N(l) = 1$ in the case of point 2.

[13] As an example of the fractal parameter calculations, Figure 1 shows the fracture trace network mapped from a 10 m $\times$ 10 m outcrop of sandstone (sample number 4 in our study; see below) by Odling and Webman [1991]. Figure 2 shows the corresponding $\log N(r) - \log r$ plot and $\log \Sigma N(l) - \log l$ plots. Notwithstanding the limitations noted above, the plots in Figure 2 reveal linear (fractal) behavior over a range of 1 m $< r < 10$ m and 1.3 m $< l < 4.5$ m. The fractal parameters D , a , and l_{max} of the fracture network pattern can be determined straightforwardly from these plots.

[14] In the present study, fracture network patterns have been analyzed at scales ranging from 6.5 cm $\times$ 6.5 cm to 100 km $\times$ 100 km. Figure 3 and Table 1 show the localities of the sampling points and the domain sizes, respectively, in our study region. Figure 4 shows the straight-line representations of the sampled fracture network patterns and their associated $\log N(r) - \log r$ and $\log \Sigma N(l) - \log l$ plots. The log-log plots reveal that these fracture patterns exhibit fractal spatial and length distributions, parameterized by D and a , l_{max} , respectively.

2.2. A Random Binary Fractal Fracture Network (RBFFN)

[15] In order to study fracture network connectivity, we have performed a Monte Carlo simulation using a random binary fractal fracture network (RBFFN) model characterized by the three fractal parameters D , a , and l_{max} . The

Table 2. Parameters Used in the Analysis for Percolation in the Random Binary Fractal Fracture Network Models

Parameter	Value
D	1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0
a	0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0, 2.2, 2.4, 2.6, 2.8, 3.0
l_{max}/L	0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0
Θ_1 , deg	normal distribution mean value = 90, SD = 7.5
Θ_2 , deg	normal distribution mean value = 180, SD = 7.5
m	6
J	10

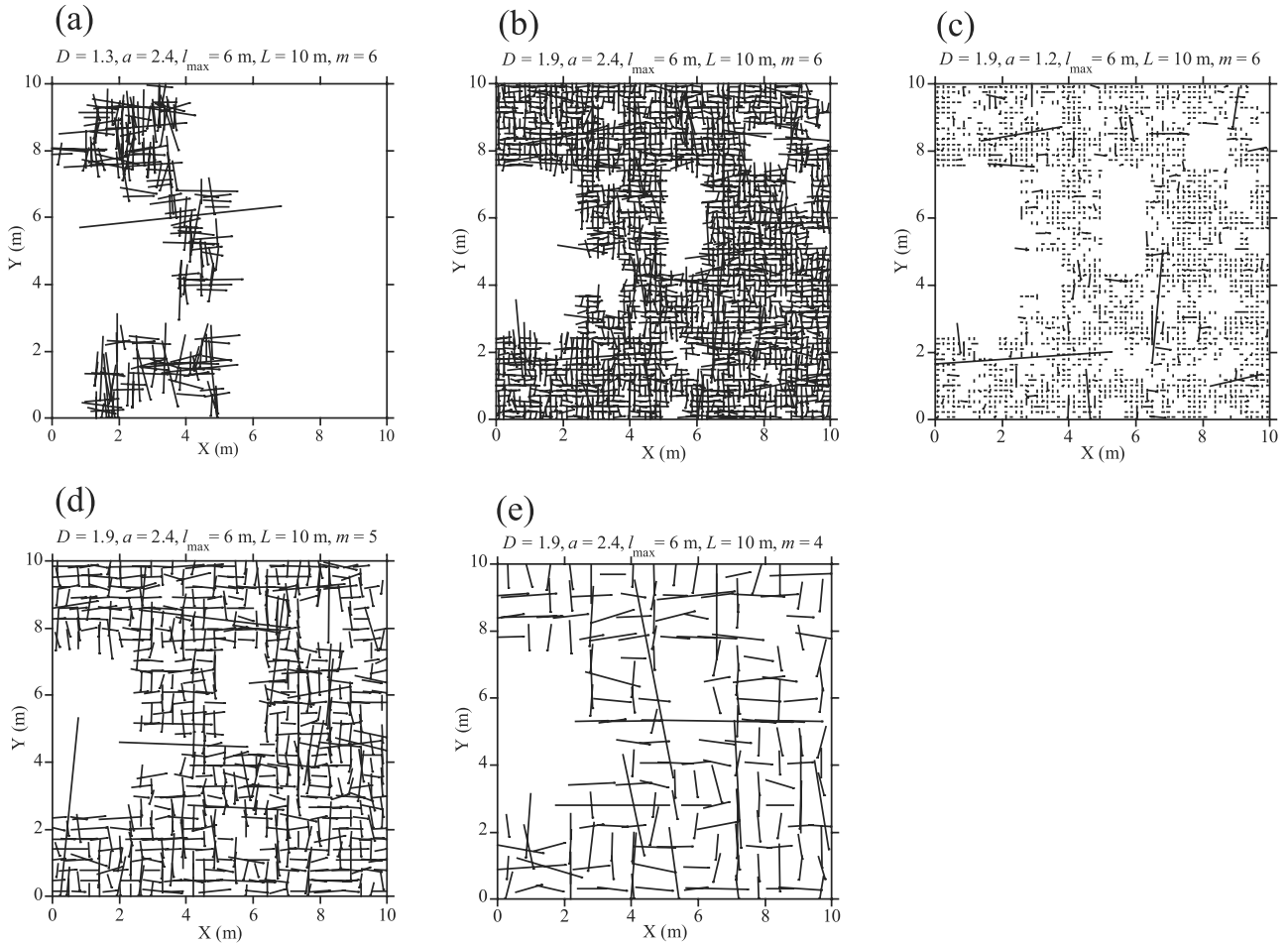


Figure 6. The examples of a random binary fractal fracture network (RBFFN) model. The parameters are (a) $D = 1.3$, $a = 2.4$, $l_{\max} = 6$ m, $L = 10$ m, $m = 6$, (b) $D = 1.9$, $a = 2.4$, $l_{\max} = 6$ m, $L = 10$ m, $m = 6$, (c) $D = 1.9$, $a = 1.2$, $l_{\max} = 6$ m, $L = 10$ m, $m = 6$, (d) $D = 1.9$, $a = 2.4$, $l_{\max} = 6$ m, $L = 10$ m, $m = 5$, and (e) $D = 1.9$, $a = 2.4$, $l_{\max} = 6$ m, $L = 10$ m, $m = 4$.

RBFFN is generated by assigning fracture centers, lengths, and orientations to fractures on a two-dimensional plane.

[16] Field observations often reveal that fracture sets have recognizable preferred orientations rather than purely random orientations. Commonly, fractures networks contain two or three fracture sets, the orientation of each of which can be represented by a distributional form such as a spherical normal (Fischer) distribution [Mardia, 1972]. Robinson [1983, 1984] has found that the average number of intersections per fracture at percolation is independent of any anisotropy of the orientations. In this study, fracture orientations are randomly chosen from two orthogonal, normally distributed fracture sets to obtain a primitive, representative two-dimensional fracture model.

[17] We proceed as follows: first, the fracture positions are generated according to the value of D using the “fractal fragmentation” method, illustrated in Figure 5a. In a flow domain of dimensions $L \times L$, the fragmentation algorithm selects 2^{Dk} of the 2^{2k} boxes in the k th iteration ($k = 1, 2, 3, \dots, m$). For example, when $k = 1$ and $D = 1.4$, three ($2^{1.4 \times 1}$) of the four ($2^{2 \times 1}$) boxes of size $r = (1/2)L$ are chosen randomly. This procedure is repeated for $k = 1$ to m steps, but only those boxes that are selected during

one iteration are available for selection in successive iterations. The spatial distribution of boxes chosen in this manner agrees with the previously determined fractal dimension to within the rounding error of the original box-counting calculation. The second step is to identify all boxes chosen during the final ($k = m$) iteration as fracture centers, and to assign lengths l_i and orientations Θ_i according to a and $l_{\max}$ and the fracture set normal distributions respectively, where $i = 1, 2, 3, \dots, 2^{Dm}$ (Figure 5b). Fractures are permitted to cut each other without interaction in this fracture network model.

3. Analysis

[18] The relationship between the percolation probability and the three fractal parameters, D , a , and $l_{\max}$ is analyzed with the Monte Carlo method. Percolation through the RBFFN model occurs when fractures connect opposite sides of the $L \times L$ flow domain. Once the first fracture path across the block has formed, the number of potential paths quickly increases with higher a and D values. Therefore the percolation probability, P , is calculated according to $P = J_p/J$, where J is the RBFFN model

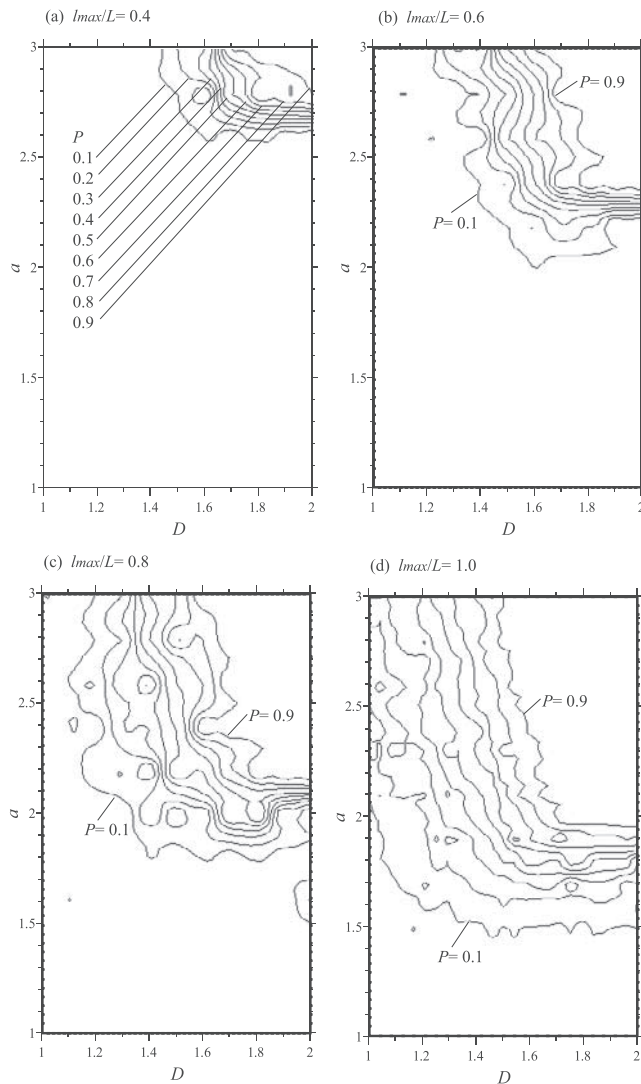


Figure 7. Contours of percolation probability, P at four different $l_{\max}/L$ on D versus a plot.

iteration number for the same parameter set (D , a and $l_{\max}/L$), and J_p is the frequency with which percolation has occurred in J iterations. In this study, the simulation was performed with $m = 6$ (Figure 5) and $J = 10$. Table 2 shows the parameters used in the analysis, and Figure 6 illustrates an example of the RBFFN model. On the basis of our natural fracture network observations, a and D are limited to 0.6–3.0 and 1.0–2.0, respectively, and $l_{\max}/L$ is limited to 0–1. The examples shown in Figures 6a–6c reveal the effects of a and D for constant $l_{\max}/L = 0.6$. Higher values of D produce greater numbers of fractures, whereas higher values of a produce an overall lengthening of the fracture set. Several intact blocks that are devoid of fractures appear spontaneously in the models. Such architecture is often seen in natural fracture networks (see Figures 1, 3, and 4). Smaller fractures tend to form clusters because the fractal fragmentation extends over a small region (of $(1/2^6)L \times (1/2^6)L$) in the flow domain (Figure 5a). The fracture skeleton that is chiefly formed by the long fractures is apparently obscured by many small

fractures of greater m value in the RBFFN model, as shown in Figures 6b, 6d, and 6e. However, the fracture skeleton in the RBFFN model is maintained even if the m value increases from 4 to 6 with the same fractal parameters. Figure 6 also suggests that the number of small fractures not belonging to the percolation cluster increases as a decreases.

4. Results and Discussions

4.1. Percolation Conditions for Two-Dimensional Fracture Networks

[19] The results of the Monte Carlo RBFFN simulations are shown in Figure 7 as contours of percolation probability, P , at different values of $l_{\max}/L$ on the $D - a$ plot. P changes continuously with the three fractal parameters, D , a , and $l_{\max}/L$, but the results indicate that percolation seldom occurs when either $0.1 \leq l_{\max}/L \leq 0.3$ and $a \leq 3.0$, or $0.6 \leq a \leq 1.0$. We define the percolation threshold (the boundary between percolating and nonpercolating behavior) as the $P = 0.5$ contour on the $D - a$ plots (illustrated schematically in Figure 8).

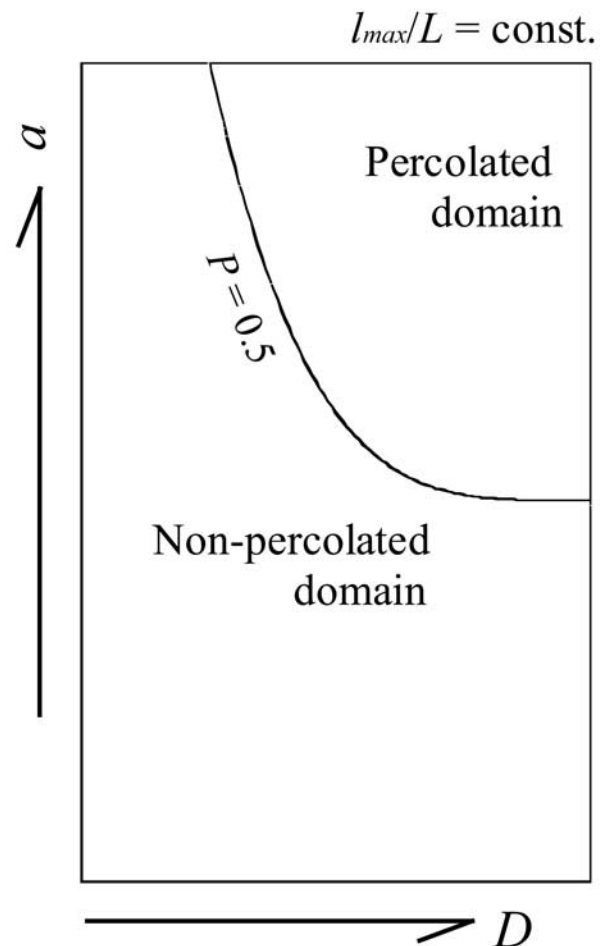


Figure 8. Schematics of the boundary between the percolated and nonpercolated domain for D versus a relation at constant $l_{\max}/L$.

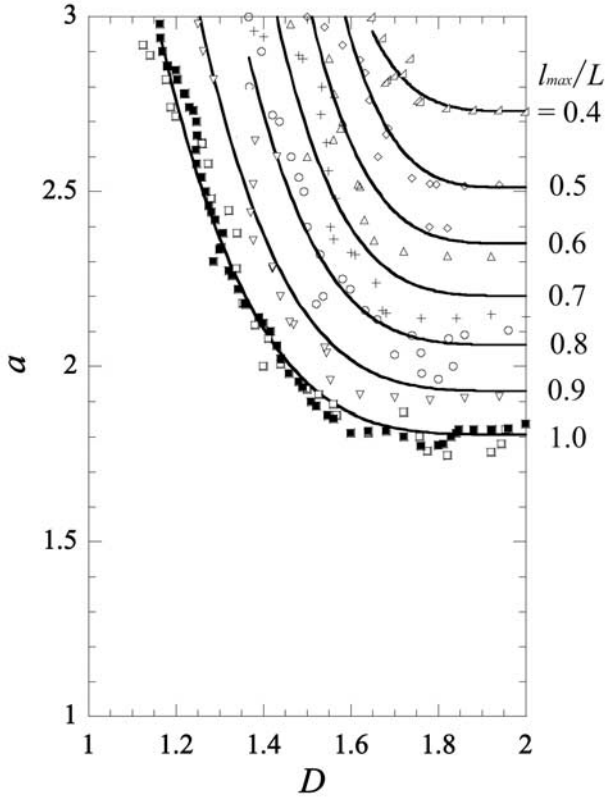


Figure 9. Numerical results for D versus a relationships at $P = 0.5$ in different $l_{\max}/L$ with the best fitting curves that form into equation (3), $a = m_1(L')(D - 2)^4 + m_2(L')$, ($D < 2$, $1 < a < 3$, $L' < 1$); $m_1(L') = 117.5e^{-3.926L'}$; $m_2(L') = 3.492e^{-0.6582L'}$; $L' = l_{\max}/L$. Equation (3) provides with the boundary between the percolated and nonpercolated domain for fracture networks according to a binary fractal.

[20] Figure 9 shows the percolation threshold for different values of $l_{\max}/L$ and the best fitting general relationship of the form

$$a = m_1(L')(D - 2)^4 + m_2(L'), \quad (D < 2, 1 < a < 3, L' < 1) \quad (3)$$

where

$$m_1(L') = 117.5e^{-3.926L'},$$

$$m_2(L') = 3.492e^{-0.6582L'},$$

$$L' = l_{\max}/L.$$

The relationships between $m_1(L')$ or $m_2(L')$ and L' of equation (3) were determined by fitting the data shown in Figure 10 with exponential curves. Equation (3) describes the boundary between the percolated and nonpercolated domains for the RBFNs and can therefore be used to predict whether percolation will occur through two-dimensional fracture networks of the form considered here for arbitrary values of the three fractal parameters D , a and $l_{\max}/L$.

[21] Bour and Davy [1997] described a percolation threshold relationship in terms of P and the fracture density

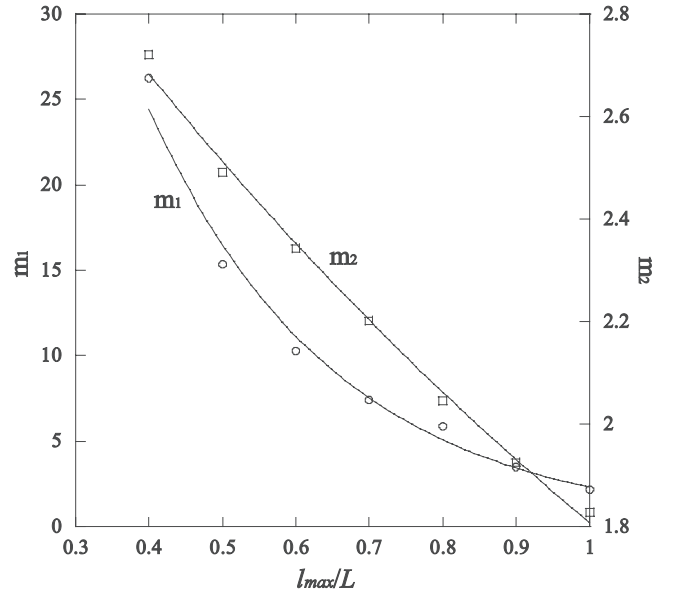


Figure 10. Relationship between $m_1(L')$ or $m_2(L')$ and L' ($=l_{\max}/L$) of equation (3).

(cumulative fracture length per unit area) based on percolation theory applied to random fault networks possessing power law fault length distributions. Here, we have also examined the role of fracture density using the binary fractal approach. The apparent fracture density, ρ' , of a fracture network pattern corresponding to a binary fractal is expressed, based on equation (2), as follows:

$$\rho' = \sum_{i=1}^N L_i/L^2 = (1/L)(l_{\max}/L) \sum_{i=1}^{N(l)} (i)^a, \quad (4)$$

where $N(l) = (2^k)^D$.

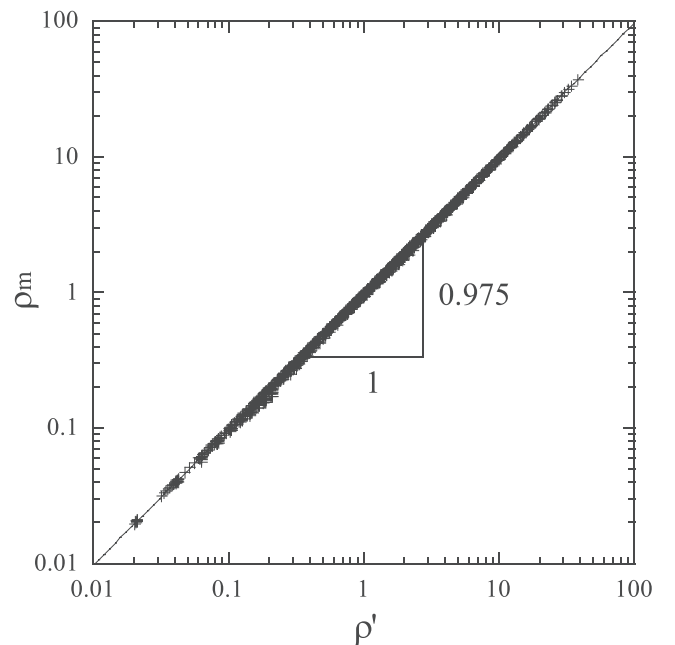


Figure 11. Relationship between ρ' and ρ_m .

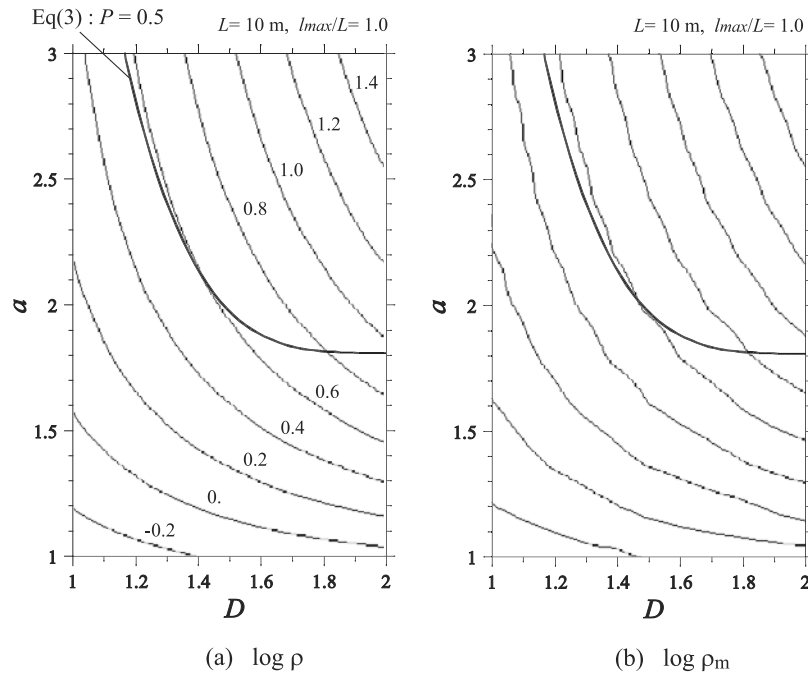


Figure 12. Contours of (a) $\log \rho$ and (b) $\log \rho_m$ with equation (3) at $l_{\max}/L = 1.0$ on D versus a plot.

[22] However, equation (4) overestimates the fracture density because it incorporates fractures extending beyond the $L \times L$ domain. Comparing the fracture density described by equation (4) with the mean value ρ_m determined using the RBFFN model, enables us to adjust ρ' to account for this limitation. Figure 11 illustrates the relationship between ρ' and ρ_m for different values of D , a , and $l_{\max}/L$, with $L = 10$ m and $k = 6$. The data reveal a linear relationship between the logarithms, namely $\log \rho_m = 0.975 \log \rho'$. Therefore the true fracture density, ρ , can be straightforwardly determined from equation (4) and with an adjustment factor of 0.975:

$$\log \rho = 0.975 \log \rho', \quad (5)$$

Figure 12 shows $\log \rho$ and $\log \rho_m$ as functions of the three fractal parameters (D , a , and $l_{\max}/L$) as expressed by equation (3); ρ coincides very well with ρ_m , but equation (3) is discordant with the $\log \rho$ contour. This disagreement between equation (3) and the isodensity line means that P cannot be simply expressed as a function of ρ . Therefore, though equation (3) is an empirical rather than theoretical relationship, it is presumed to better represent the percolation threshold than the fracture density-based method. This is presumably because it is based on measurable fractal parameters.

4.2. Application of the Percolation Threshold Relationship to Natural Fracture Networks

[23] It is clear from Figures 2 and 4 that the fracture networks of samples 1–4 lie within the percolation domain (in the sense that fractures extend completely across each sample), while samples 5–7 are in the nonpercolation domain. In order to confirm the viability of the numerical modeling approach to hydraulic conductivity estimation, predictions of percolation in the natural fracture network patterns were made by comparing the measured D , a , and

$l_{\max}/L$ values with those predicted using equation (3) to be necessary for percolation (Figure 8). Figure 13 and Table 3 show the measured parameters and the boundary between the percolated and nonpercolated domains predicted by equation (3) for each fracture sample. These results substantiate the use of equation (3) in predicting whether a two-dimensional fracture network pattern in crystalline rocks enables fluids to percolate: in two-dimensional fracture networks with $l_{\max}/L < 1.0$ and $a < 1.8$, percolation seldom occurs irrespective of D (< 2) because of the network's poor hydraulic connectivity. Most natural fracture networks, in which $a < 3$, are unlikely to percolate for $l_{\max}/L < 0.25$.

[24] We can also apply equation (3) to earthquake catalog data to investigate relationships between fluid migration and seismic activity. Byerlee [1993], Miller *et al.* [1996], and Yamashita [1997] concluded based on theoretical results and numerical simulations that fluid migration plays an important role in seismicity. Pervasive fluid migration and fluctuations in water quality appeared in association with the 1965–1966 Matsuhiro earthquake swarm, Japan [e.g., Iijima, 1969; Tsuneishi and Nakamura, 1970], and Tsunogai and Wakita [1995] reported precursory water quality changes associated with the 1995 Hyogoken-Nanbu (Kobe) earthquake ($M = 7.2$), which they interpreted as the results of fluid migration from deep fracture zones. Whether the complexity of seismic phenomena is in part related to fluid redistribution processes is an intriguing scientific problem. If fluid migrates in response to seismic events, then the faults responsible for the earthquakes must be connected to each other in a percolating network. Many workers have estimated the fractal dimension D and multifractal dimension of earthquake catalogs from hypocenter distributions [e.g., Kagan and Knopoff, 1980; Kagan, 1981a, 1981b; Geilikman *et al.*, 1990; Hirata and Imoto, 1991; Hirabayashi *et al.*, 1992; De Rubeis *et al.*, 1993; Godano and Caruso, 1995; Legrand *et al.*, 1996; Godano *et al.*,

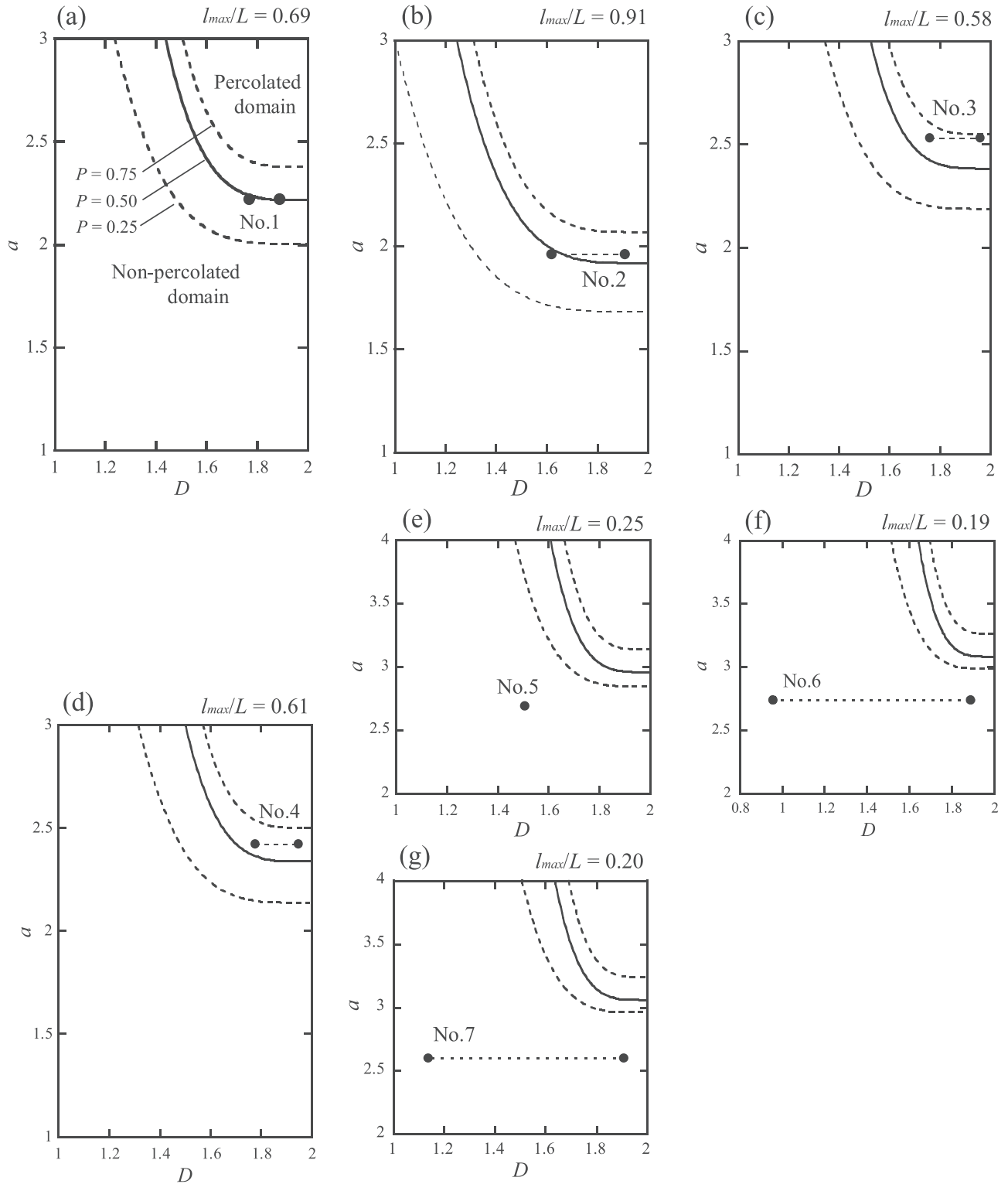


Figure 13. Measured fractal parameter (solid circle) and predicted boundary (solid line; $P = 0.5$) between the percolated and nonpercolated domain for D versus a relation at different $l_{\max}/L$ for natural fracture patterns of samples 1 to 7. The dashed lines show the $P = 0.25$ and $P = 0.75$.

1997; Nakaya and Hashimoto, 2002]. The Gutenberg-Richter formula, which relates earthquake magnitude M and cumulative frequency N , is of the form $\log N = a_m - bM$ and can be recast as a $\log N - \log l$ relationship if an appropriate

empirical relationship between fracture size S and magnitude exists. One such relationship is $\log S = M - 4.07$ [Sato, 1979], and the fractal parameter a can be obtained directly in terms of the b value of the Gutenberg-Richter relation-

Table 3. Comparison of the Measured and the Predicted by Means of equation (3) for Percolation in Samples of the Natural Fracture Network Patterns

No.	D	a	$l_{\max}$, m	$l_{\max}/L$	Measured	Predicted
1	1.77–1.89	2.22	0.045	0.69	percolated	percolated
2	1.62–1.91	1.96	0.68	0.91	percolated	percolated
3	1.76–1.96	2.53	3.50	0.58	percolated	percolated
4 ^a	1.78–1.95	2.42	6.10	0.61	percolated	percolated
5 ^b	1.51	2.69	12,500	0.25	nonpercolated	nonpercolated
6 ^b	0.956–1.89	2.74	9,450	0.19	nonpercolated	nonpercolated
7 ^b	1.14–1.91	2.60	19,700	0.20	nonpercolated	nonpercolated

^aFractures are traced from *Odling and Webman* [1991].^bFaults are traced from *Research Group for Active Faults in Japan* [1991].

ship, as $a = 2b$. The b value varies from region to region but is generally in the range $0.8 < b < 1.2$ [Frohlich and Davis, 1993]. Even within a single region, the b value varies temporally in the range $0.6 < b < 1.6$ over long periods [e.g., Hirata, 1989a; Smirnov, 1995; Wyss et al., 1997; Murase and Iizuka, 1999]. Then $a = 2$ for a typical b value of 1, and inserting this value and $l_{\max}/L = 1$ into equation (3) yields the relationship $D > 1.5$ as the percolation threshold condition. Therefore, under the condition $D = D_3 - 1$, where $D_3 \geq 2.5$ is the value of D in three-dimensional space and the b value is 1, the dynamic fracture network exceeds the percolation threshold. In order to apply this sort of analysis to real seismicity, it will be necessary to measure three-dimensional fractal parameters. Since fracture sampling is usually carried out on a scan plane or scan line such as a tunnel wall, field outcrop, or borehole, it is also necessary to clarify the relationship between two- and three-dimensional fractal parameters [e.g., La Pointe et al., 1993]. In hot dry rock power generation projects, artificial fractures are typically created by hydraulic fracturing in order to construct artificial reservoirs. In a similar manner to that used for naturally occurring earthquakes, we could construct a seismic catalog containing the origin time, latitude and longitude, depth, and magnitude of low magnitude seismic events induced during hydraulic fracturing [e.g., Sato and Fujii, 1989; Main, 1991]. These data may enable the fractal parameters of the fracture network, and hence the network's overall connectivity, to be determined, possibly as functions of time.

5. Summary

[25] In this study, seven natural rock fracture patterns at various scales were mapped. On the basis of the geometric characteristics of each network, a random binary fractal fracture network (RBFFN) model was developed to numerically simulate percolation probability with the Monte Carlo method. The binary fractal is described by three parameters, which characterize the spatial distribution of fractures (D), the distribution of fracture lengths (a), and the maximum fracture length normalized by the domain length ($l_{\max}/L$). The percolation probability of a fracture network varies continuously with these parameters, so we have developed a percolation threshold condition that describes mathematically the boundary between percolating and nonpercolating domains. The numerical results indicate that applying the percolation condition to natural fracture networks measured

at various scales substantiates the hypothesis that the three fractal geometric parameters indeed control the fracture connectivity and hence the migration of fluids through rock fractures, and active and seismogenic faults. Our numerical approach may be equally applicable to studies of earthquake-fluid interaction based on seismicity catalogs.

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